## AD 2. AERODROMES

### VOGA AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOGA - MANOHAR INTERNATIONAL AIRPORT, MOPA, GOA

### VOGA AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

| | | | |
|---|---|---|---|
| 1 | Aerodrome reference point coordinates and its site | 154433N 0735201E<br>North of TWY A, East of RET A4 abeam TWY W | |
| 2 | Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves | 16.5 km North-East of Mapusa city | |
| 3 | Aerodrome elevation and reference temperature | 564 FT / 34.6 DEG C | |
| 4 | Magnetic variation, date of information and annual change | 1.50 DEG W (2010) /0.033 DEG E | |
| 5 | Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available) | GMR Goa International Airport Limited (GGIAL), MOPA, Pernem, North Goa, Goa-403512,<br>Goa - 403512 | |
| | | Telephone: | (91) 832 2408444 |
| | | Fax: | |
| | | AFS: | |
| | | Email: | airsidemgr.ggial@gmargroup.in<br>Website- www.miagoaairport.com |
| 6 | Types of traffic permitted (IFR/VFR) | IFR/VFR | |
| 7 | Remarks | 1. Aerodrome Reference Code: 4E.<br>2. Vertical Reference System: EGM-08<br>3. Non-Scheduled flight operators intending to operate to Manohar International Airport, Mopa, Goa are required to file their slot request with slots.ggial@gmrgroup.in and aocc.ggial@gmrgroup.in for clearance.<br>4. All Non-scheduled / General Aviation aircraft will be subject to positive approval from Airport Operator.<br>5. Request to be forwarded by flight operator for operation of International GA movements at least 48 hrs in advance and for Domestic GA movements at least 24 hrs in advance.<br>6. For NSOP aircraft, night parking not permitted for more than 24hrs. | |

### VOGA AD 2.3 OPERATIONAL HOURS

| | | |
|---|---|---|
| 1 | Aerodrome Operator | MON to FRI and 1st and 3rd SAT: 0400-1230 UTC (0930-1800 IST)<br>All other days: NIL |
| 2 | Custom and immigration | H24 |
| 3 | Health and sanitation | H24 |
| 4 | AIS briefing office | H24 |
| 5 | ATS reporting office (ARO) | H24 |
| 6 | MET Briefing office | H24 |
| 7 | Air Traffic Service | H24 |
| 8 | Fuelling | H24 |
| 9 | Handling | H24 |
| 10 | Security | H24 |
| 11 | De-icing | NIL |

| | | |
|---|---|---|
| 12 | Remarks | NIL |

**VOGA AD 2.4 HANDLING SERVICES AND FACILITIES**

| | | |
|---|---|---|
| 1 | Cargo-handling facilities | Available |
| 2 | Fuel and Oil types | JET A1 |
| 3 | Fuelling facilities and capacity | Fueling agency – BPCL;<br>Fueling with Fuel bowsers & Hydrant<br>Total Capacity: 7500KL |
| 4 | De-icing facilities | NIL |
| 5 | Hangar space for visiting aircraft | NIL |
| 6 | Repair facilities for visiting aircraft | NIL |
| 7 | Remarks | NIL |

**VOGA AD 2.5 PASSENGER FACILITIES**

| | | |
|---|---|---|
| 1 | Hotel(s) at or in the vicinity of aerodrome | In the city. |
| 2 | Restaurant(s) at or in the vicinity of aerodrome | At Airport and in the city. |
| 3 | Transportation possibilities | Taxies & Buses. |
| 4 | Medical Facilities | At Airport |
| 5 | Bank and post office at or in the vicinity of aerodrome | Banks: ATMs at Airport<br>Post office: In the city |
| 6 | Tourist office | At Airport |
| 7 | Remarks | NIL |

**VOGA AD 2.6 RESCUE AND FIRE FIGHTING SERVICES**

| | | |
|---|---|---|
| 1 | Aerodrome category for fire fighting | Within ATS HR: CAT-9 |
| 2 | Rescue equipment | Available as per the Category |
| 3 | Capability for removal of disabled aircraft | 1. The critical aircraft for Manohar International Airport, MOPA, GOA is B777-300ER.<br>2. Primary responsibility for removal of disabled aircraft rests with the concerned airline.<br>3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit.<br>4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations:<br>Aerodrome Coordinator:<br>Head-ARFF<br>Mob: 9156888332<br>Email: arffep.ggial@gmrgroup.in<br>5. The capability of Disabled Aircraft removal available up to Beechcraft-200 with station arrangement. |
| 4 | Remarks | NIL |

**VOGA AD 2.7 SEASONAL AVAILABILITY CLEARING**

| | | |
|---|---|---|
| 1 | Type(s) of clearing equipment | NIL |
| 2 | Clearance priorities | NIL |
| 3 | Remarks | NIL |

## VOGA AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

| | | |
|---|---|---|
| 1 | Designation, surface and strength of aprons | Refer Aircraft Parking / Docking Chart |
| 2 | Designation, width, surface and strength of taxiways | Refer Aircraft Parking / Docking Chart |
| 3 | Location and elevation of altimeter checkpoints | Location : All Aircraft Stands<br>Elevation: Refer AD 2.23 for details |
| 4 | Location of VOR checkpoints | On TWY A1 and TWY A6 |
| 5 | Position of INS checkpoints | Not Available |
| 6 | Remarks | Refer AD 2. 23 for additional information |

## VOGA AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

| | | |
|---|---|---|
| 1 | Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands | Illuminated stand identification boards displaying stand number and stand co-ordinates are available for stands 4, 5, 6, 7A, 7, 7B, 8A, 8, 8B, 9, 10, 11, 12, 13,14A, 14, 14B, 15A, 15, 15B, 50 and 51.<br><br>Taxiing guidance signs are provided on all TWY intersections and straight sections.<br><br>Advanced Visual Docking Guidance System installed on stands 8A, 8, 8B, 9, 10, 11, 12 and 13. Aircraft to follow AVDGS for parking.<br><br>Marshalling guidance available for parking in assigned stands through Authorized Ground Handling Agents/Airlines. |
| 2 | Runway and taxiway markings and lights | RWY Marking: Runway Transverse stripe, Runway Designation, Runway Threshold, Runway Touchdown zone, Runway Centreline, Runway Aiming Point, Runway Side stripe markings.<br><br>RWY Lights: Runway Threshold, Runway Edge, Runway Centreline and Runway End lights.<br><br>TWY Marking: Taxiway Edge, Taxiway Centre line and Runway Holding positions, Intermediate holding position (on intersections of TWY A with A2,A3, A5, TWY W, TWY E, intersections of Taxilane D with TWY W and TWY E)and Enhanced Taxiway centre line markings.<br><br>TWY Lights : Blue in colour, omnidirectional taxiway edge Lights are available at all taxiways.<br><br>Runway guard lights configuration ‘A’ available at TWYs A1, A2, A3, A4, A5 A6 and Y.<br><br>No Entry bar lights provided for RET A3 and A4. |
| 3 | Stop bars (if any) | Not Available |
| 4 | Remarks | Mandatory & Information signs provided.. |

## VOGA AD 2.10 AERODROME OBSTACLES

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM 08) | Marking/LGT | Remarks |
| 28/TKOF 10/APCH | OTHER | 154449.9N 0735035.5E | 511 FT | NIL | Mobile road traffic (road elev. + traffic height 150.70+5) |
| In circling area and at AD | BUILDING | 154440.2N 0735225.3E | 568 FT | Marked | GP Hut |
| In circling area and at AD | NAVAID | 154439.9N 0735228.3E | 573 FT | Marked/LGTD | GP Monitor Antenna |
| In circling area and at AD | NAVAID | 154440.2N 0735225.4E | 578 FT | LGTD | GP DME |
| In circling area and at AD | NAVAID | 154440.1N 0735225.4E | 611 FT | Marked/LGTD | GP Antenna |
| In circling area and at AD | ANTENNA | 154440.1N 0735225.4E | 616 FT | NIL | Arrester on GPAntenna |
| In circling area and at AD | ANTENNA | 154447.0N 0735053.4E | 544 FT | Marked/LGTD | IMD Antenna |
| In circling area and at AD | ANTENNA | 154439.9N 0735223.2E | 592 FT | Marked/LGTD | IMD Antenna |
| In circling area and at AD | TREE | 154517.8N 0735915.7E | 1139 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154913.1N 0735639.5E | 1146 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154920.5N 0735631.9E | 1151 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154930.9N 0735455.2E | 1227 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154655.3N 0735901.2E | 1382 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154935.6N 0735609.6E | 1399 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154540.2N 0735923.4E | 1494 FT | NIL | Group of Trees on Hill Top |
| In circling area and at AD | TREE | 154950.3N 0735610.9E | 1518 FT | NIL | Group of Trees on Hill Top |

## VOGA AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

| | | |
|---|---|---|
| 1 | Name of the associated meteorological office | AMS |
| 2 | Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours | H24 |
| 3 | Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts | AMO Mumbai |
| 4 | Availability of the trend forecast for the aerodrome and interval of issuance | Trend<br>Forecast appended to METAR |
| 5 | Information on how briefing and/or consultation is provided | Self-briefing displayThrough Online Briefing System (OLBS) |
| 6 | Types of flight documentation supplied and language(s) used in flight documentation | Wind/temperature charts of different flight levels and sig. Weather charts as available in OLBS.<br>English |

| | | |
|---|---|---|
| 7 | Charts and other information displayed or available for briefing or consultation | Charts available in OLBS |
| 8 | Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images; | Displays installed at ATC tower, Met Briefing Room and Approach Radar |
| 9 | The air traffic services unit(s) provided with meteorological information | Manohar International Airport ATC and ACS |
| 10 | Additional information, e.g. concerning any limitation of service. | Aviation Weather Observation Systems available for both Runways 10 & 28.<br>Integrated panel available at Met Briefing Room and Control Tower.<br>Instrumental RVR reading for Runway 10 & 28 available through Present Visibility Sensor. |

## VOGA AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations | TRUE Bearings | Dimensions of RWY (M) | Strength of pavement (PCR) and associated data) and surface of runway and associated stopways | Geographical coordinates for threshold and runway end |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 10 | 94.68 DEG | 3500 x 45 M | | THR:<br>154444.51N<br>0735040.26E<br>RWY END:<br>154435.19N<br>0735237.47E |
| 28 | 274.68 DEG | 3500 x 45 M | | THR:<br>154435.19N<br>0735237.47E<br>RWY END:<br>154444.51N<br>0735040.26E |

| THR elevation and highest elevation of TDZ of precision APP RWY | Slope of runway and associated stopway | Dimensions of stopway (M) | Dimensions of clearway (M) | Dimensions of strips (M) |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| THR: 503.8FT<br>TDZ: 524.6FT | 0.52% | NIL | NIL | 3620 x 280 M |
| THR: 563.9FT<br>TDZ: 563.9FT | -0.52% | NIL | NIL | 3620 x 280 M |

| Dimensions of runway end safety areas | Location and description of arresting system (if any) | Existence of an obstacle-free zone | Remarks. |
|---|---|---|---|
| 11 | 12 | 13 | 14 |
| 240M x 90M | NIL | AVBL | Vertical Reference System: EGM-08<br>PCR RWY 10: 1021/F/A/W/T |
| 240M x 90M | NIL | AVBL | Vertical Reference System: EGM-08<br>PCR RWY 28: 1021/F/A/W/T |

## VOGA AD 2.13 DECLARED DISTANCES

| RWY Designator | Take-off run available TORA (M) | Take-off distance available TODA (M) | Accelerate distance available ASDA (M) | Landing distance available LDA (M) | Remarks (including runway entry or start point where alternative reduced declared distances have been declared) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 10 | 3500 | 3500 | 3500 | 3500 | NIL |
| 28 | 3500 | 3500 | 3500 | 3500 | NIL |

## VOGA AD 2.14 APPROACH AND RUNWAY LIGHTING

| Runway Designator | Type, length and intensity of approach lighting system | Runway threshold lights, colour and wing bars | Type of visual slope indicator system | Length of runway touchdown zone lights |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| **10** | SALS<br>420 M<br>LIH | Green | PAPI<br>LEFT/3.00 DEG<br>MEHT<br>(66.73FT) | NIL |
| **28** | CAT I<br>900 M<br>LIH | Green | PAPI<br>RIGHT/3.00 DEG<br>MEHT<br>(66.73FT) | NIL |

| Length, spacing, colour and intensity of runway centre line lights | Length, spacing, colour and intensity of runway edge lights | Colour of runway end lights and wing bars | Length and colour of stopway lights | Remarks |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| 3500 M<br>30 M<br>LIH | 3500 M<br>60 M<br>White | Red | | 1. RWY centreline lights White from the threshold to the point 900 m from the runway end, alternate red and white from 900m to 300 m from the runway end and red from 300 m to the runway end.<br>2. The colour of runway edge lights in last 600m is yellow |
| 3500 M<br>30 M<br>LIH | 3500 M<br>60 M<br>White | Red | | 1. RWY centreline lights White from the threshold to the point 900 m from the runway end, alternate red and white from 900 m to 300 m from the runway end and red from 300 m to the runway end.<br>2. The colour of runway edge lights in last 600m is yellow<br>3. Sequence flashing lights supplementing ALS available. |

## VOGA AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

| | | | |
|---|---|---|---|
| 1 | Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any) | ABN | Top of the control tower<br>- Green and white flashing lights. 24 FPM/12 RPM. |
| | | IBN | Not Available |
| 2 | Location and lighting (if any) of anemometer/ landing direction indicator; | LDI | Landing T provided between RWY and TWY A in between TWY A3 and TWY A4 and illuminated. |
| | | Anemometer | Anemometer Located within the Met park at both ends of RWY and near ATC building.<br><br>WDI provided at beginning and end of Runway on the north sides.<br><br>All Wind Direction Indicator are illuminated. |
| 3 | Taxiway edge and taxiway centre line lights; | Edge | Taxiway edge Lights are available for TWYs A, A1, A2, A3, A4, A5, A6, D, E, W, Y and Z.<br><br>Spacing 60m at straight sections and 15m at curves. |
| | | Centre Line | Not Available |
| 4 | Secondary power supply including switch-over time; | All ground lights are connected to UPS ensuring 1 second of switch over time.<br><br>Under all operating conditions including LVP condition, GED power through Online 100KVA UPS shall be primary source and 3 X 2MVA diesel generator (designated at MRSS) power will be secondary source ensuring less than 1 sec switch over time. | |
| 5 | Remarks | NIL | |

## VOGA AD 2.16 HELICOPTER LANDING AREA

| | | |
|---|---|---|
| 1 | Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area | Not established |
| 2 | TLOF and/or FATO area elevation: | Not established |
| 3 | TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking; | Not established |
| 4 | True bearings of FATO; | Not established |
| 5 | Declared distances available | Not established |
| 6 | Approach and FATO lighting; | Not established |
| 7 | Remarks | Not established |

## VOGA AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

| | | |
|---|---|---|
| 1 | Airspace designation, geographical coordinates and lateral limits | CTR: Area bounded by lines joining points 155237N 0733119E; 155237N 0741620E; 153855N 0741620E; 153545N 0741131E; 153545N 0733119E to point of origin. |
| 2 | Vertical limits | FL 85 |
| 3 | Airspace classification | D |
| 4 | Call sign and language(s) of the air traffic services unit providing service; | Mopa Tower / Mopa Approach, English |
| 5 | Transition altitude | 6000 FT |
| 6 | Hours of applicability | HO |

| | | |
|---|---|---|
| 7 | Remarks | NIL |

## VOGA AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

| Service Designation | Call sign | Channel(s) | SATVOICE Number(s), if available |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| APP | Mopa Approach | 123.800 MHZ | |
| APP | Mopa Approach | 125.150 MHZ | |
| TWR | Mopa Tower | 118.725 MHZ | |
| TWR | Mopa Tower | 119.400 MHZ | |
| ATIS | Mopa Information | 127.675 MHZ | |
| ALRS | EMERG FRQ | 121.500 MHZ | |
| SMC | Mopa Ground | 118.725 MHZ | |
| SMC | Mopa Ground | 121.750 MHZ | |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| 5 | 6 | 7 |
| | H24 | Main |
| | H24 | Backup |
| | H24 | Backup |
| | H24 | Main |
| | H24 | NIL |
| | H24 | NIL |
| | H24 | Backup |
| | H24 | Main |

## VOGA AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid | Identification | Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate | Hours of operation, as appropriate; |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| LOC 28 | IMPA | 110.500 MHz | H24 |
| GP 28 | | 329.600 MHz | H24 |
| DME ILS 28 | IMPA | CH42X | H24 |
| DVOR/DME | MPG | 117.400 MHz CH121X | H24 |

| Geographical coordinates of the position of the transmitting antenna | Elevation of transmitting antenna of DME/ elevation of GBAS reference point | Service volume radius from the GBAS reference point | Remarks |
|---|---|---|---|
| 5 | 6 | 7 | 8 |
| 154445.4N 0735029.5E | | | |
| 154440.1N 0735225.4E | | | |
| 154440.2N 0735225.4E | 578 FT | | LP DME |
| 154448.9N 0735124.0E | 563 FT | | HP DME |

**VOGA AD 2.20 LOCAL AERODROME REGULATIONS**

DOM NON SKED OPR TO OBTAIN 24HR PRIOR APPROVAL AND INTL NON SKED OPR TO OBTAIN 48HR PRIOR APPROVAL BFR CONDUCT OF FLT.CTC AP OPR ON 919156999114 AND 919156999119 FOR APPROVAL

**VOGA AD 2.21 NOISE ABATEMENT PROCEDURES**

Consistent with safety of aircraft operations pilots are encouraged to minimize the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome. Pilots are also encouraged to use single engine for taxiing in/out consistent with safety of aircraft operations.

**VOGA AD 2.22 FLIGHT PROCEDURES**

## I. LOW VISIBILITY PROCEDURE

### 1. INTRODUCTION:

In compliance with DGCA Civil Aviation Requirements (CAR) Section 8, Series 'C', Part-I on All-weather Operations, Low Visibility Procedures are to be enforced at Aerodromes for ensuring safe operations during Categories II and III approaches and/or low visibility take-offs. Also, in the Revision 10 to the CAR para 5.3 following provision regarding Low Visibility Procedures is added. "An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550M RVR/800M visibility unless low visibility procedures are enforced". Para 5.2 of the ibid CAR provides that "An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA. Scheduled operators may be authorized LVTO minima of up to 125 m".

This provision necessitated the establishment of Low Visibility Procedures for accommodating/permitting departures in Visibility/RVR less than 800M/550M even at such airports where there are no CAT II and CAT III operations. As per provisions of CAR Section 4 Series B Part I para 5.3.12.3 and ICAO Annex 14, Runway centre line lights are mandatory for take-off in RVR below 400 m. Further, as per para 9.8.7 of the ibid CAR, Surface Movement Radar (SMR) shall be provided at an aerodrome intended for use in Runway Visual Range conditions less than a value of 350 M.

Manohar International Airport has Runway with designation 10/28. RWY 28 is equipped with ILS and RWY 10 is non-precision. While RWY 28 is equipped with CAT-I approach lighting system (supported by Runway edge lights and Runway centre line lights with 30 m spacing and relevant RVR information), RWY 10 is equipped with simple approach lighting system. Airport is not equipped with SMR and ASMGCS.

Accordingly, Low visibility Procedures have been developed for Manohar Airport to accommodate/permit departures in Visibility/RVR less than 800M/550M but not lower than 350 m from RWY 10/28. Whenever instrument RVR is not available, manual RVR observation is required subject to availability of required facilities.

**2. DEFINITIONS:**

**2.1** Non-precision Approach Runway: A runway served by visual aids and a non-visual aid intended for landing operations following an instrument approach operation type A and a visibility not less than 1000 m.

**2.2** Precision Approach Runway Category I (CAT I) operation: A precision instrument approach and landing with a decision height not lower than 60m (200 feet) and with either a visibility not less than 800m, or a runway visual range not less than 550 meters.

**2.5** Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: as per Para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550M RVR/800M Visibility unless low visibility procedures are enforced.

**2.6** Low Visibility Take Off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m

**2.7** Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

**2.8** Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

**2.9** Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a. Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.
b. Landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and
c. Landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

**2.10** Visibility - Visibility for aeronautical purposes is the greater of:

a. the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;
b. the greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

**2.11** Runway Visual Range (RVR) – An instrumentally derived value based upon standard calibrations that represent the horizontal distance a pilot will see down the runway from the approach end.

a. Touchdown RVR – The RVR visibility readout values obtained from RVR equipment serving the runway touchdown zone.
b. Mid-RVR – The RVR visibility readout values obtained from RVR equipment located midfield of the runway.
c. Rollout RVR – The RVR visibility readout values obtained from RVR equipment located nearest the rollout end of the runway.

**3. ABBREVIATIONS**:

AGL - Aerodrome Ground Lights

ASMGCS - Advance Surface Movement Guidance Control System

ATC - Air Traffic Control

ATCO - Air Traffic Control Officer

CAR - Civil Aviation Requirements

CFT - Crash Fire Tender

DA - Decision Altitude

DG - Diesel Generator

DGCA - Director General of Civil Aviation

DH - Decision Height

GGIAL - GMR Goa International Airport Ltd.

ICAO - International Civil Aviation Organization

LVP - Low Visibility Procedures

LVTO - Low Visibility Take Off

RVR - Runway Visual Range

SFL - Sequence Flashing Lights

SMR - Surface Movement Radar

SP - Safeguarding Procedures

SQUID - Squitter Identification

UPS - Uninterrupted Power Supply

**4. GENERAL:**

**4.1** The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate Runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

**4.2** When RVR is less than 550 m , landings are not permitted at Manohar International Airport, since aerodrome is certified only for CAT-I ILS operations. LVTO will be permitted up to RVR 350 m.

**4.3** All Aerodrome Ground Lights at Manohar International Airport are supported by UPS (Uninterrupted Power Supply) to ensure change over time of less than one second. However, during any unserviceability of the UPS when LVP in force, secondary power supply (DG) shall be switched on for supporting the AGL to ensure change over time of less than one second

**5. FACILITIES, SERVICES AND EQUIPMENT:**

**5.1** Minimum Required Facilities/Lighting System: Following minimum required facilities/lighting system are available at Manohar International Airport for supporting Low Visibility Operations:

| Facility/Lighting | Runway 10 | Runway 28 |
|---|---|---|
| Runway Edge lights | Available | Available |
| Runway Centreline lights | Available | Available |
| Runway End lights | Available | Available |
| Surface Movement Radar | Not Available | Not Available |
| ASMGCS | Not Available (only marking and lighting available) | Not Available (only marking and lighting available) |
| Present Visibility Sensor | Available for providing real-time touchdown RVR | Available for providing real-time touchdown RVR |

| | |
|---|---|
| UPS/DG sets | Available with switch over time of less than1 sec for RWY edge lights, RWY end lights & RWY threshold lights. |

**5.2** Additional Facilities Available: Following additional facilities are available at Manohar International Airport as enhanced means for supporting Low Visibility Operations:

| Facility/Lighting | Runway 10 | Runway 28 |
|---|---|---|
| No Entry bar lights | Available for RET A3 and A4 | Available for RET A3 and A4 |
| Runway guard lights | Available at all RETs and TWYs connecting Runway | Available at all RETs and TWYs connecting Runway |
| Taxiway edge lights | Available | Available |

**5.3** Effect of Unserviceability of Aeronautical lights/Equipment: Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in Para 5.1 above is un-serviceable or is not maintained as per the required standard. Post implementation, LVP operations will be suspended if any of the below mentioned conditions develops:

| AGL Facility | Un-serviceability | Effect / Restrictions / Remarks |
|---|---|---|
| Runway Edge lights | More than 15% of all lights are unserviceable. | Suspend LVTO. |
| Runway End lights | More than 15% of all lights are unserviceable. | Suspend LVTO. |
| Runway centre line lights | More than 15% of all lights are unserviceable. | No impact / restrictions for LVTO. |

Note: When any two consecutive lights are unserviceable in any of the significant elements, the system is deemed to be on outage and LVP operations will be suspended.

**6. SAFEGUARDING PROCEDURES:**

Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations in order that LVPs are implemented after all Safeguarding Procedures are completed.

**6.1 Implementation of Safeguarding Procedures** : Duty Officer Tower/Tower Supervisor will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

Safeguarding Procedures shall be initiated:

a. When the Visibility/RVR is less than 1200 m and visibility/RVR is forecast to deteriorate to 800 m/550 m or less; and/or

b. The cloud ceiling is less than 400 ft and forecast to fall to 200 ft or less.

c. Safeguarding Procedure shall be completed before RVR reaches below 800 m or before implementation of LVP.

**6.2** Actions by various units during Safeguarding Procedures

a. Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area.

b. The appropriate aeronautical ground lights must have been inspected duringsafeguarding procedure. Once AGL inspection for safeguarding procedure initiation is completed, the inspection shall be valid for 02 hours irrespective of implementation/withdrawal of LVP within this time frame or inspection to be carried out when deemed necessary.

c. Ensuring availability and readiness of secondary power supply for change over time of maximum one second for Runway edge and Runway end lights supported by UPS/DG sets.

d. RWY edge and RWY end lights may continue to operate on main power supply during Safeguarding Procedures. Whenever, LVP is to be implemented the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented.

e. The TWR controller shall ensure that the Sequence Flashing Lights (SFL) associated with the approach light shall be switched ON.

f. ln case of non-availability of real-time RVR or if the RVR instrument is unserviceable, suitable arrangements shall be made for obtaining Manual RVR for the Runway in use. Tower Supervisor shall coordinate with MET for the provision of manual RVR. Duty manager Airside Ops shall provide necessary transport/VHF facilities to MET official for this purpose.

**7. LOW VISIBILITY PROCEDURES:**

**7.1** Implementation of Low Visibility Procedures:

Whenever Visibility/RVR reduces to 800 m/550m or below and/or cloud ceiling is at 200 feet or below, Tower Supervisor shall coordinate with Airside Operations team and ARFF to confirm whether the Safeguarding Procedures have been completed. Upon confirmation of completion of Safeguarding Procedures, Tower supervisor will implement Low Visibility Procedures and communicate the same to Airside operations, AOCC and ARFF. No operations shall continue below 550 m RVR unless LVP is imposed.

**7.2** Actions by various units during LVP:

a. Duty MET Officer shall keep Duty Officer Tower/Tower Supervisor informed of any change in Visibility/RVR.

b. The following shall be included in DATIS. "LOW VISIBILITY PROCEDURES IN FORCE"

c. ATC may restrict flight operations in such a way that not more than 2 aircraft are permitted to pushback at a time and 03 aircraft on the manoeuvring area. This can be achieved by actual position reporting by the pilots and/or Follow Me.

d. ARFF shall ensure positioning of 01 CFT at pre-determined position(PDP) at CFRS - Y road, immediately on implementation of LVP..

e. TWR should preferably permit departures only from the beginning of the Runway in use.

f. ATC shall not permit any ground run on the manoeuvring area except idle power run up on the stands, after coordination with Duty Manager-Airside Operations/Follow Me.

g. All non-sked aircraft shall be provided with Follow-Me service. Other aircraft may be provided with Follow Me service on request from ATC/pilot.

h. The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all-operating on the manoeuvring area shall be maintained by Airside Operations.

i. Vehicles fitted with a serviceable VHF R/T shall only be allowed to operate on the manoeuvring area. ln case of extreme urgency, a vehicle without a VHF RT may be permitted provided it is escorted by a Follow Me Vehicle with a serviceable VHF R/T.

j. Airside Operations team shall ensure that towing of aircraft is done under escort of "Follow Me" vehicles. Only one towing at time shall be permitted.

k. Shift ln-charge Electrical shall continuously monitor the main and Standby Power supply to ensure change over time of maximum one second for RWY edge and RWY end lights during low visibility operations and report any unserviceability to Tower immediately.

l . Ensure that the RWY lights are inspected every 02 hours from the time of implementation of LVP during its continuance.

**7.3** Suspension of Aircraft Operations:

Duty Officer Tower/Tower Supervisor shall stop all flight operations whenever RVR falls below 350 m.

**8. TERMINATION OF LOW VISIBILITY PROCEDURES:**

**8.1** When Visibility/RVR improves to 800m/550M or more and cloud ceiling is 200 feet or Higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty Met Officer regarding improvement in weather conditions before the termination of LVP.

**8.2** The Tower Supervisor will intimate ARFF and AOCC regarding the termination of LVP operations.

**8.3** On termination of LVP, following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES TERMINATED”.

**8.4** Termination of Safeguarding Procedures: when visibility/RVR improves and is more than 1500m and/or the cloud ceiling is 400ft or higher Tower Supervisor/Duty Officer Tower may cancel safeguarding procedure..

**9. DUTIES AND RESPONSIBILITIES OF OTHER AGENCIES**

9.1 Before commencement of monsoon/winter season, a meeting will be held by HeadAirside Operations, GGIAL every year to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

9.2 All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations.

9.3 A refresher program for ATCO's and personnel responsible for airside operations shall be conducted every year.

9.4 All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

9.5 All the vehicles must have their obstruction lights "ON" during Low Visibility Procedures operations.

**II. RADIO COMMUNICATION FAILURE PROCEDURE**

**1. INTRODUCTION:**

Radio communication failure procedures are described in Para 15.3 of ICAO PANSATM DOC 4444. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft at Manohar International Airport, MOPA, GOA.

**2. GENERAL:**

2.1. All Transponder equipped aircraft experiencing Radio Communication failure shall set transponder to Mode A/C code 7600 as soon as practicable.

*(Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot’s decision to set transponder to Mode A/C code 7500 or 7700, whenever required).*

**3. ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:**

3.1. In case of arriving aircraft, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as assigned runway. In case arriving aircraft has not been advised any runway, Runway 28 shall be considered as assigned runway for such arrival.

3.2 In case of departures from Mopa returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

3.3 If Radio communication failure occurs after the final/interception turn, the aircraft may continue its approach to land on such Runway.

3.4 Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

3.5 Irrespective of visibility/weather conditions, Runway and Approach lights in 'SWITCHED OFF' position shall indicate non- availability of such Runway for aircraft experiencing RCF. In such cases, alternate runway shall be made available for such aircraft.

3.6 In strong tail wind conditions for assigned Runway, aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach for such approach, aircraft will carry out published ILS/ VOR DME ARC approach procedure for the runway which is in opposite direction to the assigned Runway. For example, Runway 10 in case of Runway 28 and vice versa.

**4. PROCEDURE FOR ARRIVALS:**

4.1 Pilots of RNAV1 equipped Arrivals shall;

4.1.1 When Arrival clearance has been received from ATC follow the RCF procedure corresponding to that STAR as mentioned in para 4.1.3 & 4.1.4

4.1.2 When Arrival clearance has not been received from ATC:

a) Follow the appropriate RNAV STAR corresponding to that route, except for W101W (GONAB Transition) and W15 (KANTI Transition), thereafter follow the respective RCF procedure as given in 4.1.3 & 4.1.4

b) For aircraft routing via W101W, W15 from south of the Manohar International airport, MOPA follow the RCF procedure as given in 4.1.5 & 4.1.6

**4.1.3 ARRIVALROUTE–SPECIFIC RCF PROCEDURES FOR RUNWAY 28**

At the end of the STAR, make a convenient turn to intercept localizer/final approach track of the published instrument approach procedure, to carry out the approach procedure.

**W15/V20** (UDULO Transition)

Maintain the last assigned level if,

a) at or below FL160, follow UDULO1A arrival with lateral and vertical restrictions.

b) above FL160, proceed to UDULO, descend in UDULO hold to FL160 and follow UDULO1A arrival with lateral and vertical restrictions.

**W67/J5** (EPKOS Transition)

Maintain the last assigned level if,

a) at or below FL090, follow EPKOS 1A arrival with lateral and vertical restrictions.

b) above FL090, proceed to GA555, descend in GA555 hold to FL090 and follow EPKOS 1A arrival with lateral and vertical restrictions.

**W120/J104/V19** (BBM Transition)

Maintain the last assigned level if,

a) at or below FL110, follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

b) above FL110, proceed to BBM, descend in BBM hold to FL110 and follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

**W101W** (GONAB Transition)

a) Follow GONAB 1A arrival with lateral and vertical restrictions till GA517, from GA517 proceed to R022/22D MPG hold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**W15** (KANTI Transition)

a) Follow KANTI 1A arrival with lateral and vertical restrictions till GA517, from GA517 proceed to R022/22D MPGhold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**T9/V19**

Maintain the last assigned level if,

a) at or below FL090, follow OMDEV 1A arrival with lateral and vertical restrictions.

b) above FL090, follow OMDEV 1A arrival with lateral and vertical restrictions till GA201, from GA201 proceed to R022/22D MPG hold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**4.1.4 ARRIVAL ROUTE–SPECIFIC RCF PROCEDURES FOR RUNWAY 10**

At the end of the STAR, make a convenient turn to intercept final approach track of the published instrument approach procedure, to carry out the approach procedure.

**W15/V20** (UDULO Transition)

Maintain the last assigned level if,

a) at or below FL090, follow UDULO1C arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

b) above FL090, proceed to UDULO, descend in UDULO hold to FL090 and follow UDULO 1C arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

**W67/J5** (EPKOS Transition)

Maintain the last assigned level if,

a) at or below FL120, follow EPKOS 1C arrival with lateral and vertical restrictions.

b) above FL120, proceed to GA555, descend in GA555 hold to FL120 and follow EPKOS 1C arrival with lateral and vertical restrictions.

**W120/J104/V19** (BBM Transition)

Maintain the last assigned level if,

a) at or below FL120, follow BBM 1C arrival with lateral and vertical restrictions.

b) above FL120, proceed to BBM, descend in BBM hold to FL120 and follow BBM 1C arrival with lateral and vertical restrictions.

**W101W** (GONAB Transition)

Maintain the last assigned level if,

a) at or below FL100, follow GONAB 1C arrival with lateral and vertical restrictions.

b) above FL100, follow GONAB 1C arrival with lateral and vertical restrictions till GA705, from GA705 proceed to R022/22D MPG hold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**W15** (KANTI Transition)

Maintain the last assigned level if,

a) at or below FL100, follow KANTI 1C arrival with lateral and vertical restrictions.

b) above FL100, follow KANTI 1C arrival with lateral and vertical restrictions till GA705, from GA705 proceed to R022/22D MPG hold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**T9/V19**

Maintain the last assigned level if,

a) at or below 6000FT, follow OMDEV 1C arrival with lateral and vertical restrictions.

b) above 6000FT, follow OMDEV 1C arrival with lateral and vertical restrictions till GA532, from GA532 proceed to R022/22D MPG hold, descend in R022/22D MPG hold to 4600FT and carry out the approach procedure.

**4.1.5 AIRWAY W101W AND W15 (FROM SOUTH) – SPECIFIC RCF PROCEDURES FOR RUNWAY 28, WHEN ARRIVAL CLEARANCE HAS NOT BEEN RECEIVED FROM ATC**

At the end of the STAR, make a convenient turn to intercept localizer/final approach track of the published instrument approach procedure, to carry out the approach procedure.

**W101W**

Aircraft shall route via W101W GONAB W17 BBM (to avoid VOR44B),

Maintain the last assigned level if,

a) at or below FL110, follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

b) above FL110, proceed to BBM, descend in BBM hold to FL110 and follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

**W15 (from South)**

Aircraft shall route via W15 MML W17 BBM (to avoid VOR44B),

Maintain the last assigned level if,

a) at or below FL110, follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

b) above FL110, proceed to BBM, descend in BBM hold to FL110 and follow BBM 1A arrival with lateral and vertical restrictions, cross 25 DME MPG at or above FL75.

**4.1.6 AIRWAY W101W AND W15 (FROM SOUTH)–SPECIFIC RCF PROCEDURES FOR RUNWAY 10, WHEN ARRIVAL CLEARANCE HAS NOT BEEN RECEIVED FROM ATC**

At the end of the STAR, make a convenient turn to intercept final approach track of the published instrument approach procedure, to carry out the approach procedure.

**W101W**

Aircraft shall route via W101W GONAB W17 BBM (to avoid VOR44B),

Maintain the last assigned level if,

a) at or below FL120, follow BBM 1C arrival with lateral and vertical restrictions.

b) above FL120, proceed to BBM, descend in BBM hold to FL120 and follow BBM 1C arrival with lateral and vertical restrictions.

**W15 (from South)**

Aircraft shall route via W15 MML W17 BBM (to avoid VOR44B),

Maintain the last assigned level if,

a)at or below FL120, follow BBM 1C arrival with lateral and vertical restrictions.

b) above FL120, proceed to BBM, descend in BBM hold to FL120 and follow BBM 1C arrival with lateral and vertical restrictions.

**4.1.7 ARRIVING AIRCRAFT BEING RADAR VECTORED**

Aircraft being radar vectored for approach, on experiencing RCF, shall maintain last

assigned level and heading for 3 minutes after detecting RCF or selecting MODE A/C 7600, whichever is later (If below FL75, climb and reach FL75 before crossing 25NM outbound from MPG VOR) and then take a convenient turn to proceed direct to 'MPG VOR maintaining last assigned level or FL75 whichever is higher, From MPG proceed outbound on R-022 for R022/22D hold. Descend to 4600FT in the hold and then carryout published instrument approach procedure for arrival runway.

**4.1.8 Procedure for Non-RNAV arrivals**

4.1.8.1 If RCF takes place at or beyond 50NM of MPG, continue on ATS route, (Rejoin ATS route if given heading or flying offset) descending to cleared flight level or FL 75 whichever is higher. From 'MPG VOR', proceed outbound on R-022 for R022/22D hold. Descend to 4600FT in the hold and then carryout published instrument approach procedure.

4.1.8.2 If RCF takes place within 50NM of MPG, then aircraft shall maintain last assigned level and heading for 3 minutes after detecting RCF or selecting code 7600, whichever is later (If below FL75, climb and reach FL75 before crossing 25NM outbound from MPG VOR) and then proceed direct to MPG VOR maintaining last assigned level or FL75 whichever is higher. From 'MPG VOR', proceed outbound on R-022 for R022/22D hold. Descend to 4600FT in the hold and then carryout published instrument approach procedure.

**5. PROCEDURE FOR DEPARTURE (BOTH RNAV AND NON RNAV)**

**5.1 DEPARTURE INTENDING TO CONTINUE TO DESTINATION**

Any departing aircraft experiencing RCF and intending to continue to its filed plan destination shall:

5.1.1 Continue on assigned SID or heading climbing to or maintaining cleared level or FL75, whichever is higher.

5.1.2 Three minutes after setting Mode A/C code 7600 or reaching FL 75 or cleared Flight level (if higher than FL75) whichever is later:

i) If following 'SID', climb to filed Flight Level with SID lateral and vertical restrictions to join ATS route and continue as per the filed flight plan.

ii) If following Radar heading, turn to join ATS route by shortest route maintaining cleared flight level/FL 75 whichever is higher. After joining ATS route, climb to filed Flight level and continue as per the filed flight plan.

**5.2 DEPARTURE INTENDING TO LAND BACK AT MANOHAR INTERNATIONAL AIRPORT MOPA**.

Any departing aircraft experiencing RCF after departure and intending to land back at Manohar International Airport, Mopa shall:

5.2.1 Continue on assigned SID or heading, climbing to or maintaining cleared level or FL75 whichever is higher.

5.2.2 Three minutes after setting Mode A/C code 7600 or reaching FL75 or cleared Flight level (if higher than FL75) whichever is later, take a turn to proceed direct to MPG VOR.

5.2.3 From 'MPG VOR', proceed outbound on R-022 for R022/22D hold.

5.2.4 If required to burn fuel, then remain in the hold.

5.2.5 Descend to 4600FT in the hold and then carryout published instrument approach procedure.

III. ATS Surveillance services are provided H24 to suitably equipped aircraft on opportunity basis within the coverage area of MOPA ADS-B sensors in the jurisdiction of MOPA Approach control. SER may be available beyond such time subject to availabity of manpower .ADS-B equipped aircraft flying within MOPA Approach Control to comply with relevant provisions

**VOGA AD 2.23 ADDITIONAL INFORMATION**

| 1. Preferred Exit Taxiways: | | | | |
|---|---|---|---|---|
| **RWY Designation** | **Aircraft Type** | **Preferred Exit TWY** | **Distance from THR (M)** | **Remarks** |
| 10 | Code A, B, C | A4 (RET) | 1900 | Not compatible for Code D/E |
| | Code D, E | A5 | 3055 | RET vacation not permitted |
| 28 | Code A, B, C | A3 (RET) | 1900 | Not compatible for Code D/E |
| | Code D, E | A2 | 2650 | RET vacation not permitted |

| 2. Distance of Taxiways from Runway Threshold: | | |
|---|---|---|
| **FROM THRESHOLD** | **TAXIWAY** | **DISTANCE (M)** |
| RWY 10 | A2 | 850 |
| RWY 10 | A3 | 1600 |
| RWY 10 | A4 | 1900 |
| RWY 10 | A5 | 3055 |
| RWY 10 | Y (for IAPP) | 2754 |
| RWY 28 | A5 | 445 |
| RWY 28 | Y (for IAPP) | 746 |
| RWY 28 | A4 | 1600 |
| RWY 28 | A3 | 1900 |
| RWY 28 | A2 | 2650 |

**3. Ground Handling Arrangements:**

All operators, operating to/from Manohar International Airport, MOPA, GOA Mopa shall employ the services of Ground Handling Agent authorized by GGIAL. Operators should contact the Manohar International Airport, MOPA, GOA for further information on ground handling arrangements. All Non-Scheduled operators operating to/from VOGA to make necessary ground handling arrangements with GHA's in advance.The Ground Handling Agent at Manohar International Airport, MOPA, GOA is Celebi Airport Service India Pvt. Ltd; contact details - 9004899993

| 4. Aircraft stands: | | | | |
|---|---|---|---|---|
| **Aircraft Stand** | **Coordinates** | **PCR** | **Elevation (ft) EGM-08** | **Compatibility** |
| 4 | 154358.37N 0735218.20E | 1175/R/A/W/T | 535 | All Code A, B and C type of aircraft. |
| 5 | 154358.49N 0735216.69E | 1175/R/A/W/T | 534 | All Code A, B and C type of aircraft. |
| 6 | 154358.61N 0735215.18E | 1175/R/A/W/T | 535 | All Code A, B and C type of aircraft. |
| 7A | 154359.059N 0735213.70E | 1175/R/A/W/T | 534 | All Code C type of aircraft. |
| 7 | 154358.24N 0735212.78E | 1175/R/A/W/T | 535 | All Code D & E type of aircraft. |
| 7B | 154358.13N 0735212.11E | 1175/R/A/W/T | 535 | All Code C type of aircraft. |

| | | | | |
|---|---|---|---|---|
| 8A | 154359.29N<br>0735210.69E | 1175/R/A/W/T | 534 | All Code C type of aircraft. |
| 8 | 154358.48N<br>0735209.76E | 1175/R/A/W/T | 535 | All Code D & E type of aircraft. |
| 8B | 154358.37N<br>0735209.10E | 1175/R/A/W/T | 535 | All Code C type of aircraft. |
| 9 | 154359.20N<br>0735207.65E | 1012/R/A/W/T | 535 | All Code C type of aircraft. |
| 9W | 154358.77N<br>0735207.56E | 1012/R/A/W/T | 535 | B787-800, 787-900, A330-300 & A350-900 only.<br>Docking under Follow Me guidance |
| 10 | 154359.32N<br>0735206.14E | 869/R/A/W/T | 535 | All Code C type of aircraft. |
| 11 | 154359.45N<br>0735204.64E | 869/R/A/W/T | 535 | All Code C type of aircraft. |
| 12 | 154359.56N<br>0735203.13E | 869/R/A/W/T | 534 | All Code C aircraft |
| 13 | 154359.68N<br>0735201.62E | 869/R/A/W/T | 535 | All Code C aircraft. |
| 14A | 154359.80N<br>0735200.11E | 1175/R/A/W/T | 535 | All Code A, B and C aircraft. |
| 14 | 154359.32N<br>0735159.21E | 1175/R/A/W/T | 535 | All Code D & E aircraft. |
| 14B | 154359.93N<br>0735158.61E | 1175/R/A/W/T | 535 | All Code A, B and C aircraft. |
| 15A | 154400.05N<br>0735157.10E | 1175/R/A/W/T | 535 | All Code A, B and C aircraft. |
| 15 | 154359.57N<br>0735156.20E | 1175/R/A/W/T | 535 | All Code D & E aircraft. |
| 15B | 154400.17N<br>0735155.60E | 1175/R/A/W/T | 535 | All Code A, B and C aircraft. |
| 50 | 154405.17N<br>0735227.12E | 869/R/A/W/T | 540 | All Code A, B and C type of aircraft. |
| 51 | 154405.29N<br>0735225.61E | 869/R/A/W/T | 539 | All Code A, B and C type of aircraft. |
| IAPP | 154447.70N<br>0735213.34E | 1175/R/A/W/T | 559 | Aircraft type up to and including Code E aircraft |

(a) All Parking Stands are power in/push back and the pattern of Taxi Lead-in lines require nose wheel steering while taxiing into the stand.

(b) Wide body stand lead-in lines are marked with continuous line and narrow body stand lead-in lines are marked with broken lines.

(c) Pilots should exercise caution when manoeuvring on the apron due to the proximity of other aircraft, ground staff and equipment. Engine power should be restricted to the minimum required, to reduce the adverse effect of jet blast.

(d) Marshalling service is required on all the stands to park the aircraft except stands 8A, 8, 8B, 9, 10, 11, 12 and 13 (AVDGS installed).

(e) For docking to stand 9W aircraft to follow mandatory Follow Me service due non-availability of stand ID sign.

(f) Helicopters will be parked on any of the remote parking stands. Air taxi of helicopters not permitted in the Apron except, for those helicopters which are not capable of taxiing on ground.

(g) Isolated Aircraft Parking Position capable of parking aircraft type up to and including Code E aircraft under Bomb threat, Unlawful Interference or contagious medical emergencies is designed at the northern side of RWY with access through TXY 'Y'.

| 5. Parking Restrictions on Stands: | | | |
|---|---|---|---|
| Stand No. | | No Parking | |
| Occupied by Code C | Occupied by Code D/E | Code C | Code D/E |
| 7A | --- | --- | 7 |
| --- | 7 | 7A, 7B | --- |
| 7B | --- | --- | 7 |
| 8A | --- | --- | 8 |
| --- | 8 | 8A, 8B | --- |
| 8B | --- | --- | 8,9W |
| 9 | --- | --- | 9W |
| --- | 9W | 8B,9,10 | -- |
| 10 | --- | --- | 9W |
| 14A | --- | --- | 14 |
| --- | 14 | 14A,14B | --- |
| 14B | --- | --- | 14 |
| 15A | --- | --- | 15 |
| --- | 15 | 15A,15B | --- |
| 15B | --- | --- | 15 |

6. Pushback Procedures:

a) Apron is served by single Aircraft Stand Taxilane D.

b) All aircraft up to and including Code E shall be given pushback on taxilane D.

c) All pushback from Apron-2 (Stand 50 & 51) shall be facing West only irrespective of Runway in use and shall taxi out via Taxiway Z. Only one pushback at a time shall be allowed on Apron-2.

d) East facing pushback is preferred from all stands irrespective of runway in use. However, ATC may permit pushback facing west depending on real-time situation, for smooth and orderly flow of traffic.

e) Aircraft pushing back from stand 4 shall pull/push abeam stand 5 depending on the direction of pushback.

f) Aircraft pushing back from stand 14A, 14 and 14B may be advised to pull abeam stand 13 or push abeam stand 15A in case of any arrival traffic via TWY W.

g) Aircraft pushing back from 14 and 14B if facing west, shall push abeam stand 14A to keep TWY W clear for arriving aircraft to taxi in via W towards western stands.

h) Pushback from stands 15A, 15 and 15B shall be facing east only. Aircraft pushing back from 15 shall pull abeam stand 15A.

i) Aircraft pushing back from 15B shall pull abeam stand 15. Engine start approved abeam 15 only.

j) Pushback from stand 15 & 15A will be approved subject to no wide body arrival joining taxilane D via TWY W, due jet blast clearance. No restriction if Code C is taxiing.

k) Pilots are encouraged to seek one engine start at bay before requesting for pushback to reduce delay on taxilane. Pushback may be requested once one engine is stabilized and is fully ready.

l) One-engine start on idle power on contact stands are permitted only after detaching aerobridge.

7. Pushback Separation/Restrictions

a) Separation between pushback aircraft and/or docking aircraft on apron are to be maintained as mentioned in the table below: -

| Sl. No. | Pushback of Aircraft | MNM Separation | Remarks |
|---|---|---|---|
| 1 | Code A, B, C behind Code D, E | 200 m | ±5 Code ‘C’ Stands |
| 2 | Code A, B, C in front of Code D, E | 90 m | ±2 Code ‘C’ Stands |
| 3 | Code A, B, C, D, E behind Code A, B, C | 90 m | ±2 Code ‘C’ Stands |
| 4 | Code A, B, C in front Code A, B, C | 90 m | ±2 Code ‘C’ Stands |

b) Separation between aircraft docking in on apron and pushed back aircraft are to be maintained as mentioned in the table below: -

| Sl. No. | Pushback of Aircraft | MNM Separation | Remarks |
|---|---|---|---|
| 1 | Code A, B, C in front of Code A, B, C, D, E | 45 m | ±1 Code ‘C’ Stands |
| 2 | Code D, E in front of Code A, B, C, D, E | 90 m | ±2 Code ‘C’ Stands |
| 3 | Code A, B, C, D, E behind Code A, B, C | 90 m | ±2 Code ‘C’ Stands |
| 4 | Code A, B, C, D, E behind Code D, E | 135 m | ±3 Code ‘C’ Stands |

8. Taxiing Procedures:

a) Marshalling service will be provided by Ground Handling Agent for all stands not equipped with AVDGS.

b) Rapid Exit Taxiways (RETs) shall not be used for departures even though RGLs are provided.

c) Rapid Exit Taxiways (RET A3 & A4) are compatible only for aircraft upto and including Code C only. RETs shall not be used by Wide body aircraft for runway vacation.

d) RETs of opposite runway shall not be used for vacation of aircraft (for Runway 10 - do not vacate via A3 and for Runway 28 - do not vacate via A4)

e) 180 degree turn on Runway / Apron / Taxiways is not permitted. Under exceptional circumstances ATC may permit Code C or below type of aircraft to make 180 degree turn at ends of runway. Special arrangements may be made by GGIAL to facilitate aircraft taxiing out on own power from a stand and parking at locations other than the published stands due to operational requirements.

f) Pilots are encouraged to use single engine for taxi in/out consistent with safety of aircraft operations.

9. Jet Blast Abatement Procedures:

Jet Blast prone areas on Main Apron: Stands 14A, 14, 14B and 15A are identified as jet-blast prone area. All wide body aircraft while taxiing out via TWY W are requested to use minimum power due to presence of men and equipment on these stands.

10. Commissioning of ADS-B

FAC : ADS-B

FREQ: 1090 MHZ

COORD: 1544423N 073513928E

SITE ELEV : 163.25M AMSL

MAX RANGE UPTO 250NM

**VOGA AD 2.24 CHARTS RELATED TO AN AERODROME**

1. Aerodrome Chart(17 April 2025)
2. Aerodrome Chart – Hot spot(17 April 2025)
3. Aircraft Parking/Docking Chart(17 April 2025)
4. Aerodrome Obstacle Chart Type- A (Operating Limitations) RWY 10(17 April 2025)
5. Aerodrome Obstacle Chart Type- A (Operating Limitations) RWY 28(17 April 2025)
6. Aerodrome Obstacle Chart Type- B(17 April 2025)
7. Instrument Approach Chart – ILS RWY 28 (27 November 2025)
8. Instrument Approach Chart – VOR RWY 10
9. Instrument Approach Chart – RNP RWY 10
10. Coding Table - RNP Procedure RWY 10
11. Instrument Approach Chart – RNP RWY 28
12. Instrument Approach Chart - ILS W Procedure RWY 28 (17 April 2025)
13. Coding Table-Instrument Approach Chart - ILS W Procedure RWY 28 (17 April 2025)
14. Coding Table - RNP Procedure RWY 28
15. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 10
16. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 1
17. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 2
18. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 3

19. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 28
20. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 28-Coding Table 1
21. Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 28Coding Table 2
22. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 10
23. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 1
24. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 2
25. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 10-Coding Table 3
26. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 28
27. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 28-Coding Table 1
28. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 28-Coding Table 2
29. Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 28-Coding Table 3

## VOGA AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

| Procedure | Procedure Minima | VSS Penetration |
|---|---|---|
| 1 | 2 | 3 |
| | | |